

Microcosm Dynamics: an open, scored dynamic microsimulation model

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Abstract

Microcosm Dynamics is an open model that follows individual Americans through time and computes each person’s Social Security benefits year by year. From the Panel Study of Income Dynamics (PSID), it learns how people earn, marry, divorce, have children, become disabled and die, and it draws claim ages from the Social Security Administration’s published awards. Its earnings, family and disability components pass tests on held-out or external data, as do its PSID transitions once we carry them onto the Current Population Survey. We fit the whole engine on data through 2014, and it passes a holdout test against the PSID’s 2015–2019 records. Mortality and claiming run without such a test, and four of the passes, the holdout among them, rest on public amendments we adopted to a test’s terms after an earlier attempt failed. Four blind tests compare the model with DYNASIM3’s published results, and none of them uses the tested earnings, family or disability components. Under a one-point cut to cost-of-living adjustments, the model’s benefit changes land within 0.74 percentage points of DYNASIM3’s in every age group. Raising the full retirement age to 68 cuts average benefits at ages 65 to 69 in 2030 by 5.7 percent in the model and 9.9 percent in DYNASIM3. Both projections draw no earnings after 2010, apply no retirement earnings test, model no claiming response and compute benefits with Python rules over a fixed 35 computation years. A 13 percent benefit cut raises adjusted poverty at age 67 by 0.47 percentage points in the model and 2 in DYNASIM3, and each of the four minimum-benefit versions we test reaches more women and fewer men than DYNASIM3 projects. The Axiom rules engine, which encodes the statute as machine-readable rules, so far reproduces one piece of the model’s Python rules, average indexed monthly earnings over statutory computation years. We plan to add forward earnings, the earnings test and behavioral claiming to the projections and to run the full benefit rules in Axiom.

1 Introduction

Some questions about Social Security need more than a snapshot of one year: who a benefit cut reaches decades from now, or how a higher retirement age plays out across cohorts. Dynamic microsimulation answers them by aging a population of individual people year by year, drawing their earnings, marriages, disabilities, deaths and claims, and applying the program’s rules to each life. The Urban Institute’s DYNASIM, the Social Security Administration’s MINT and the Congressional Budget Office’s long-term model, CBOLT, do this for Social Security. All three run inside their institutions on code or administrative records that are not public, so outside analysts cannot rerun them or reproduce their results (Favreault et al. 2015; Smith et al. 2021; Congressional Budget Office 2018).

Anyone can read, rerun and check Microcosm Dynamics. It extends Microcosm, PolicyEngine’s open cross-sectional microdata for the United States, with a longitudinal layer we estimate on the Panel Study of Income Dynamics (PSID). We publish its code, estimates and tests under the MIT license at github.com/PolicyEngine/microcosm-dynamics.

Earnings, family formation and disability each pass a gate: a test on held-out or external data. Before a gate’s scored run, we fix the statistics it scores, which we call cells, and a tolerance for each cell. We change them only by a public amendment, which must precede the next scored run. The processes we estimate on the PSID also pass a transport gate, which tests them once we carry them onto the Current Population Survey (CPS) frame, the representative population the model can start from. Four gates pass only under such amendments, which we adopted after an earlier attempt failed (Section 4). They cover the earnings histories the model generates, family formation, the CPS transport and the whole engine, which we fit on data through 2014 and score on its 2015–2019 projections against the PSID. Mortality, claiming, disability-insurance entitlement and the immigrants who enter the population on a schedule run without a gate, and no gate certifies how mortality rates drift over time.

Four blind tests compare the model with results published from DYNASIM3, and none of them uses a gated component. In each test, we post the plan publicly before the run, and two independent readings of DYNASIM3’s published values stay sealed until we have committed the model’s output to the repository. Each plan, which we call a registration, discloses its participants’ exposures, including any to those published values, and a test is blind only as far as those disclosures show. For a one-point cut to cost-of-living adjustments, the model’s 2030 benefit changes land within 0.74 percentage points of DYNASIM3’s in every age group, and the absolute gap averages 0.38 points. Once we fix the year from which each benefit’s cost-of-living increases run, that cut acts mechanically on benefits in payment, which makes this comparison a weak test of behavioral dynamics.

Each of the other three tests finds a gap between the two models. Raising the full retirement age to 68 cuts average benefits for people aged 65 to 69 in 2030 by 5.7 percent in the model and 9.9 percent in DYNASIM3, so the model’s cut is 4.2 points shallower. A 13 percent benefit cut raises adjusted poverty at age 67 in the 1936–45 birth cohort by 0.47 percentage points in the model and 2 points in DYNASIM3. The model observes only the cohort’s odd birth years. DYNASIM3 projects that the headline version of a minimum benefit reaches 6.3 percent of beneficiaries aged 62 and older, and the model finds 7.7 percent. These last two tests apply each reform to incomes the PSID already recorded, while DYNASIM3’s figures come from projections its authors published in 2004 and 2006, so a gap in either test can reflect the population’s vintage as well as the two models’ rules.

The two projection tests, which cover the cost-of-living cut and the higher retirement age, draw no earnings after 2010. The model’s forward earnings law, which draws each person’s employment and earnings rank every two years, passes its own gate against the PSID’s 2016 and 2018 earnings, waves we had in view when we chose the law’s structure. We fit that law on PSID earnings through 2014 for ages 25 to 64. It has no fitted distribution above 64 and draws anyone older from its oldest age band. Both projection tests start in 2010 and need earnings for people in their sixties, the ages at which the retirement earnings test withholds benefits from beneficiaries who work before full retirement age. The model computes the projection tests’ benefits with Python rules. The Axiom rules engine so far reproduces one piece of those rules, average indexed monthly earnings computed with statutory computation years, while the projection tests count a fixed 35 years. We plan to connect the forward law to the projection tests, apply the retirement earnings test, add behavioral claiming and run the full benefit rules in Axiom (Section 6).

2 The model

The model fits most transitions of individual lives on the PSID, draws claim ages from the Social Security Administration’s published awards, and runs these transitions forward, one year at a time, on a population of people. It has three layers. A starting population supplies the people, the projection engine moves each person through one year (`engine/loop.py`), and the rules layer turns the simulated lives into Social Security benefits (`ss/`).

2.1 The starting population

The model starts from either Microcosm’s certified, representative frame of the United States or a closed PSID cohort. We build the frame from the Current Population Survey (CPS), and it holds 166,302 people in 57,240 households, with a 2024 reference period (`gates.yaml`, `gate_w1`). We estimate the transitions on the PSID, a multi-decade family panel, and carry them onto the CPS frame. Gate `w1`, the transport gate, tests whether those transitions reproduce the frame’s cross-sectional joint distributions of earnings, age, sex, marital status and household composition.

All four blind tests against DYNASIM3 use a closed PSID cohort (`cohorts/`). We build each comparison’s cohort with code that reads the PSID files, and we verify the variable labels that code uses. The two projection tests, which run the engine forward, start from 11,405 members of the 2011 PSID wave. Their earnings histories run from 1968 through 2010, with PSID-reported earnings where the panel has them and imputed or zero values where it does not.

2.2 The projection engine

Each simulated year, the engine takes every person’s record through eight steps in a fixed order (`engine/loop.py`): mortality, aging, the marital core (which updates marital state), fertility, disability, earnings, claiming and household composition. We build each step as a separate component that does its own calculations. The loop fixes only the order of the steps and the flow of marital state: fertility (step 4) and household composition (step 8) both read the one result the marital core (step 3) produces, so every downstream step works from the same marriage history. Scheduled immigrant cohorts enter through a dedicated entry point in the loop, and births come from the fertility step.

The engine takes every random number from a dedicated stream, keyed by draw, period and module, and by person where a step needs it (`engine/rng.py`). The same seed produces an identical projection.

We fit the engine’s estimated components only on data dated no later than a boundary year, now 2014 (`engine/refit.py`). We use the boundary to score the engine out of sample. After fitting the whole engine on data through 2014, we project across the later PSID waves and compare the projection with what the PSID recorded. Gate `m6` scores that comparison over 2015–2019 (`gates.yaml`, `gate_m6`). The model treats 2020–2022 as an exogenous shock outside its scope; we publish that window as a diagnostic and keep it out of every gated cell. We hold the fitted parameters, but not every design choice, to data through 2014. We selected gate `m6`’s scored cells with the post-2014 waves in view (`gates.yaml`, `gate_m6`).

The engine separates scored holdout runs from open forward projections (`engine/support.py`). A scored holdout run may condition on which people the PSID observed in later waves. An open forward projection, which no gate scores, gets no later-wave information: the engine refuses to supply it.

2.3 The rules layer

The rules layer computes Social Security benefits from the engine’s output. Its Python oracle, the reference code in `ss/` against which we check the Axiom rules engine, computes average indexed monthly earnings (AIME), the primary insurance amount (PIA), claiming-age adjustments, and spouse and survivor benefits under 42 U.S.C. 415 and 402. It takes its parameters from the PolicyEngine-US parameter tree. The oracle counts computation years in one of two ways (`ss/statutory_aime.py`): a fixed 35 for every worker, or the statutory number, which falls short of 35 for workers who reach 62 with fewer elapsed years.

Axiom, a rules engine that encodes the statute as machine-readable rules, reproduces the oracle’s AIME for all 7,486 people we test, using statutory computation years. Those people are the members of the 2011-wave cohort who reach 62 between 1979 and 2030. A bridge program (`axiom_benefit_bridge.py`)

passes the cohort’s stored earnings histories into the real Axiom engine. With a fixed count of 35 years instead, the oracle’s AIME differs from the statutory AIME for 251 of the 7,486, all born before 1929.

We compare Axiom with the oracle on AIME alone, and both compute it on an old-age basis that assumes no period of disability. We give Axiom and the oracle the same input definitions, and the match depends on them. Both treat PSID labor income as covered earnings, and both receive the same zero or imputed values for years before 1968, years after 2010 and years absent from the panel. Of the 320,375 earnings rows the bridge sends, 194,465 hold zeros or imputed values. Axiom and the oracle produce identical output from identical records. The match says nothing about whether either AIME equals a person’s true AIME. Axiom’s Social Security encodings await acceptance, so the blind tests that compute benefits run on the Python oracle (`cola_track_a/benefits.py`).

3 What the model covers today

None of the four blind tests uses a component that passes a gate (Table 1). Of the four blind tests, exercise 1 projects how a one-point cut to cost-of-living adjustments changes benefits in 2030, and exercise 3 does the same for a full retirement age of 68. These two are the projection tests. The other two apply a reform to incomes the PSID recorded: a uniform 13 percent benefit cut in exercise 2 and a minimum benefit in exercise 4. Each verdict we report comes from its gate’s committed result file under `runs/`. Gates 1, 2, w1 and m6 pass only under public amendments, which we adopted after earlier attempts failed (Section 4).

Table 1: What each component of the model does, its test and status, and which of the four blind tests against DYNASIM3 use it. SSDI is Social Security Disability Insurance; SSI is Supplemental Security Income.

Component	What the model does	Test and status	Used in the blind tests
Recorded earnings	PSID head and spouse labor income, 1968–2022, with unreported years imputed or set to zero	Data, with imputed gaps	Yes, in exercises 1, 3 and 4
Generated earnings histories	A backcast process that generates earnings careers	gate 1: passes against held-out panel moments (<code>runs/gate1_rank_knn_v5.json</code>)	No
Forward earnings	Rank-chain law fitted on 2014 and earlier, ages 25–64	gate m6: passes, 4 of 5 seeds (<code>runs/gate_m6_candidate3_v1.json</code>)	No
Earnings test	SSA’s exempt amounts for 2000–2026 and the statutory formula that projects later years	Tested against SSA’s published amounts; not applied	No
Mortality	Death probabilities by age band and sex	Not gated; gate m6 certifies nothing about mortality drift	Exercises 1 and 3 use SSA’s 2004 period life table; exercise 2 annuitizes wealth with NCHS 2000 tables
Disability	Work-limitation incidence, recovery, occupancy and exits near full retirement age	gate m4: passes, 5 of 5 seeds (<code>runs/gate_m4_hazard_v1.json</code>)	No

Component	What the model does	Test and status	Used in the blind tests
SSDI entitlement	Awards, recoveries and conversion at full retirement age, at rates through 2008	Not gated	Yes, in exercises 1 and 3
Claiming	SSA’s claim-age distribution of retired-worker awards (Annual Statistical Supplement, Table 6.B5.1), by sex and entitlement year	Not gated	Yes, in exercises 1 and 3, with no response to reform
Marriage and fertility	Marriage, divorce, remarriage, widowhood and births	gate 2: passes, 4 of 5 seeds (runs/gate2_hazard_v16.json)	No
Household composition	Who lives with whom	gate 2b: passes, 4 of 5 seeds (runs/gate2b_hazard_v9.json)	No
Marriage by earnings	The joint of marital status and earnings	gate 2c: passes, 4 of 5 seeds (runs/gate2c_hazard_v2.json)	No
Entrants	Scheduled immigrant cohorts; births come from the fertility step	Report-only; not gated	No
Base frame	Transport of PSID processes onto the CPS frame	gate w1: passes (runs/gate_w1_candidate4_v1.json)	No
Benefit rules	AIME, PIA, auxiliaries in the Python oracle	Reference implementation	Yes, in exercises 1, 3 and 4
Axiom execution	Statutory rules in the Axiom engine	AIME matches the oracle for 7,486 of 7,486 people	No
SSI, income and poverty	Static SSI response and an adjusted poverty measure	Registered test, exercise 2	Yes, in exercise 2, which cuts reported benefits directly
Wealth and pensions	PSID wealth annuitized into income	No projection	Exercise 2 only

Earnings. The model has three kinds of earnings (recorded, generated and projected), and the blind tests use only the recorded kind. Recorded earnings are the labor income of PSID heads and spouses for income years 1968–2022 (`data/family.py`). The PSID records them annually through 1996 and every other year afterward, and we impute unreported years or set them to zero. Roughly half of the career years behind the projection tests’ benefits hold imputed values or zeros. The oracle computes benefits directly from these recorded careers. The projection tests use careers through 2010. The minimum-benefit test, exercise 4, reads a longer span of the same panel. For the odd-numbered years from 2001 to 2021, which the biennial PSID skips, it uses the following wave’s reports.

Generated earnings come from a backcast, a backward-looking process that produces earnings careers. Gate 1 scores that process against held-out panel moments, relative to the panel’s own noise floor, and it passes (`runs/gate1_rank_knn_v5.json`).

Projected earnings come from the forward earnings law (`engine/forward_earnings.py`), a rank chain we fit on earnings the PSID observed through 2014 at ages 25 to 64. Every two years, the law draws whether each person works and where they rank in the earnings distribution, then scales that rank to the national average wage. It has no fitted distribution above 64 and draws people over 64 from its oldest age band. The engine assigns fixed zero earnings to anyone without a fitted 2014 earnings state (`engine/earnings_domain.py`).

The forward law passes gate m6 in 4 of 5 random seeds, its only certification (`runs/gate_m6_candidate3_v1.json`; Section 4). Gate m6 scores the whole engine’s holdout projection (Section 2), and its earnings cells compare persistence, growth, mobility, low earnings and the zero-earnings rate in 2016 and 2018 with the PSID’s records. We fit the law’s parameters only on data through 2014, but we selected its structure with the later waves in view (`gates.yaml`, `gate_m6`).

No blind test yet draws forward earnings, and the projection tests draw no earnings after 2010 (`cola_track_a/adapters.py`). Their projections begin in 2010, four years before the forward law’s 2014 fit boundary, and they reach ages above 64, where the law has no fitted distribution. The retirement earnings test withholds benefits from beneficiaries in their sixties who work before full retirement age. Both projection tests’ specifications list it among the differences from DYNASIM3 that they leave out, and no projection applies it yet. The model already holds the earnings test’s parameters: SSA’s exempt amounts for 2000–2026, which we checked against SSA’s published history, and the statutory formula that projects later years.

Family. Three family gates pass, each in 4 of 5 seeds. Gate 2 scores marriage, divorce, remarriage, widowhood, the stock in each marital state and births (`runs/gate2_hazard_v16.json`). Gate 2b scores household composition (`runs/gate2b_hazard_v9.json`), and gate 2c scores the joint distribution of marital status and earnings (`runs/gate2c_hazard_v2.json`). The projection tests use none of these gated family models. Their marital step sets a person’s status to widowed when a linked spouse dies in the simulation, and it draws no marriages and no divorces.

Disability, mortality and claiming. The disability module passes gate m4 in 5 of 5 seeds (`runs/gate_m4_hazard_v1.json`). It models self-reported work limitation: incidence, recovery, occupancy and exits near full retirement age. Gate m4 scores it against held-out PSID employment-status records and against the age shape and exit composition of SSA’s 2023 disability-insurance statistics. It never scores SSA’s levels. Work limitation is a different concept from a Social Security Disability Insurance (SSDI) award. The projection tests use a separate SSDI entitlement component, which draws awards, recoveries and conversion at full retirement age at rates observed through 2008 (`engine/di_entitlement.py`). No gate scores it.

The engine’s mortality step uses a table of death probabilities by age band and sex, and gate m6 certifies nothing about how mortality drifts over time. The projection tests use SSA’s 2004 period life table, which we rescale each year to match the 2008 Trustees Report. Claiming runs with no gate. The claiming step draws a claim age from SSA’s published distribution of retired-worker awards by sex and entitlement year (`claiming.py`). For entitlement years after 2008, the projection tests reuse the 2008 distribution. No claim responds to a reform. Exercise 3, which raises the full retirement age to 68, therefore holds each person’s projected claim age fixed under the reform. Its registration, the plan we post before the run, lists claiming responses only as alternatives that we report beside the main result (Section 5).

Population. Gate w1 passes (`runs/gate_w1_candidate4_v1.json`). When we carry the transitions we estimate on the PSID onto the CPS frame (Section 2), they match the frame on all 44 gated cells: 43 cross-sectional joint distributions and one ordering check. The entrant schedule lets new people enter a projection. We report its output, and no gate scores it. No fitted law takes that output as an input.

Rules. Exercises 1, 3 and 4 compute benefits with the Python oracle (`cola_track_a/benefits.py`).

The projection tests count a fixed 35 computation years for every worker. Axiom, using the statutory count instead, reproduces the oracle’s AIME for all 7,486 cohort members we test (Section 2). Exercise 2 applies no benefit rules. It cuts reported Social Security income by 13 percent inside an income concept we define in Python, applies a static Supplemental Security Income (SSI) offset and computes an adjusted poverty measure. It also annuitizes PSID wealth into income with NCHS 2000 tables.

4 How the model is scored

We admit to the model any component that passes a gate, and four components run in it without one: mortality, claiming, disability-insurance entitlement and entrants. Each gate names its coverage, the held-out data it scores against, the cells it scores and a tolerance for each cell. We write all of a gate’s terms into `gates.yaml` and fix them before the scored run. We call each version of the component under test a candidate, and we register it, give it one scored run against its gate and commit the result under `runs/` whatever the verdict. A failed candidate therefore stays in the record next to the one that passed. We change a gate’s terms only by public amendment, which must precede the next scored run.

We use four kinds of gate: holdout, anchor, transport and temporal holdout. A holdout gate scores a component on persons we excluded from its fit. The family gates hold out 20 percent of persons and score every candidate on the same five random seeds. To judge the model against the sampling variation of the data a gate scores, we set each cell’s tolerance against a noise floor: the discrepancy between two disjoint halves we draw from one real panel. A seed passes only if every gated cell stays within its tolerance, and a family gate needs at least four of its five seeds to pass. An anchor gate scores a component on a person-disjoint half of the panel and against external administrative publications. The disability gate compares the age shape and exit composition of work-limitation estimates with SSA’s 2023 disability-insurance statistics. It scores none of SSA’s levels.

The transport gate, `w1`, tests whether the processes we estimate on the PSID reproduce the CPS frame’s cross-sectional joint distributions. The temporal holdout, `m6`, fits the whole engine on data from 2014 and earlier and projects it forward. It scores projected marital flows and disability flows for 2015 to 2019, and earnings moments for 2016 and 2018, against PSID records. It publishes the 2020–2022 window as a diagnostic and scores none of it. Each gate also states what it does not certify. Gate `m6` certifies nothing about how mortality and widowhood drift over time, or about entrants.

Four gates pass only under amendments we adopted after an earlier candidate failed: gates 1, 2, `w1` and `m6`. Gate 1 passes under two amendments, and `gates.yaml` records both (`gate_1`). In a distinguishability test, a classifier tries to tell generated careers from recorded ones. The first amendment makes a failing test of this kind report-only, a status for cells we publish with no tolerance, and adds a gated block that scores the careers in benefit space, which weights earnings the way the benefit formula does. The second scores another distinguishability test by its mean over 20 seeds. The passing candidate, `runs/gate1_rank_knn_v5.json`, is identical to its failed predecessor. Gate 2 passes on its 16th candidate, a changed model, under an amendment to the gate’s estimator and a split of the gate into parts that we score separately. We adopted both after earlier candidates failed (`gates.yaml`, `gate_2`). The amendment scores the gate by its mean over 20 draws. The split leaves marriage and fertility in gate 2, and we score household composition and marriage by earnings in separate gates, 2b and 2c.

Gate `w1` passes under three amendments, each of which we adopted after a candidate failed. Together they make 21 cells report-only, and narrow the one ordering check the gate scores. As at gate 1, the passing candidate is identical to its failed predecessor (`runs/NOTES-gate_w1_candidate4.md`). At gate `m6`, the third candidate passes under one amendment, which restricts the earnings cells to people who have a fitted 2014 earnings state and re-derives the cells’ noise floors. Candidate 1 failed before that amendment and candidate 2 after it (`docs/amendments/gate_m6_amendment_1_closed_domain_floors.md`). Other gates pass as well, gate `m4` among them.

We run the blind tests against DYNASIM3 (Section 5) under the gates’ procedure, with no tolerances. Because DYNASIM3 is another published model and not ground truth, each comparison reports its gaps without grading them.

Over the long horizon, assumptions outweigh the model’s machinery, so we limit what the model claims according to how far ahead it projects. The 2025 Trustees Report puts the 75-year actuarial balance at -3.82 percent of taxable payroll on intermediate assumptions, -0.33 percent on low-cost assumptions and -8.60 percent on high-cost ones (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds 2025). That low-to-high spread, 8.27 percentage points of taxable payroll, exceeds twice the intermediate deficit. We therefore sort the model’s outputs into three tiers.

Tier 1 covers distributional reform analysis that holds the population and the assumption path fixed. Baseline and reform share much of the demographic uncertainty, so it cancels when we difference them. The tier design labels the parts that do not cancel, claiming responses among them, and anchors claiming-response ranges to quasi-experimental evidence (Mastrobuoni 2009; Behaghel and Blau 2012). Tier 2 covers near-term outputs: covered earnings, beneficiary counts, claiming ages, average benefits and disability incidence. Because administrative publications report all five every year, we can score each against those publications annually. We define gate 3 in `gates.yaml` to score this tier and have not yet run it. Tier 3 covers the long horizon. For it, the design calls for sensitivity surfaces over documented assumption ranges, computed on named assumption paths, among them the Trustees’ intermediate path, and presents none of them as the model’s own forecast.

5 Four blind tests against DYNASIM3

Four blind tests set the model’s results against DYNASIM3’s published values. Two project benefit reforms to 2030: exercise 1 cuts each cost-of-living adjustment (COLA) by one percentage point, and exercise 3 raises the full retirement age to 68 (Urban Institute 2010). Two calculate directly on incomes the PSID observed: exercise 2 cuts every benefit by 13 percent (Butrica and Uccello 2004), and exercise 4 adds a minimum benefit (Favreault et al. 2006). The COLA test matches DYNASIM3 within 0.74 percentage points in every age group. Each of the other three finds a place where the two models differ.

An interactive scorecard at <https://microcosm.institute/dynamics/scorecard> publishes every cell, each test’s timeline and every registered alternative, a variant that a test’s registration names before the run and that we report beside the test’s main result. We post every registration and every result as a public comment on issue #42 of the repository, <https://github.com/PolicyEngine/microcosm-dynamics/issues/42>.

5.1 The protocol

In each test, a separate comparison workstream, working apart from the model’s builders, makes two independent readings of DYNASIM3’s published values, transcribing them from tables or digitizing them from figures. It reconciles the two readings and seals the files under SHA-256 hashes, digital fingerprints that change if a file changes. We ratify the model specification and freeze it at a named commit, a fixed version of the code in the repository. A forecaster blind to the sealed values then predicts both the model’s result and its gap as intervals, and we hash the forecast. The registration carries the specification hash, the commit, the primary row that the test reports as its result, the registered alternatives we report beside it, and the forecast. No registered alternative may replace the primary row after the fact. The model runs and writes its output to a file, the artifact, and we score only that output. We check the artifact and commit it, and only then does the comparison lane open the seal. We publish the comparison whatever the outcome. No test has an acceptance threshold.

The hashes show that neither the sealed DYNASIM3 files nor the artifact files changed. Commit

times and the recorded step order show that we committed each output before its test’s seal opened. Neither can show whether a participant ever saw a DYNASIM3 value. Each registration discloses its participants’ exposures, and a test is blind only as far as those disclosures show. The four tests ran on 23, 25 and 27 September 2026, in the order 1, 3, 2, 4.

Table 2: Each blind test’s registration, run, commit and posted result, in the order the tests ran. All times are UTC.

Test	Registration posted (UTC)	Run	Artifact committed	Result posted
1. COLA cut	2026-09-23 23:06:06	23:06–23:31	23:40:38	23:42:53
3. Full retirement age 68	2026-09-25 22:35:30	22:36–23:07	23:27:01	23:40:59
2. 13 percent cut	2026-09-27 05:01:06	05:01:49–05:03:29	05:06:13	05:29:39
4. Minimum benefit	2026-09-27 12:18:29	12:19:17–12:19:48	12:36:32	12:58:12

5.2 The projections

Both projection tests run the same engine configuration. The engine seeds a closed cohort from the PSID’s 2011 wave and ages it to 2030 in 20 random draws, with 2008-vintage mortality and disability rates and the statutory COLA and eligibility rules. The Python oracle computes benefits with a fixed 35 computation years for every worker. The projections draw no earnings after 2010, so the oracle takes each worker’s career through 2010. For people already retired when the projection starts, it takes their observed 2010 benefit amounts, and for disabled workers it uses approximate benefit levels.

Each test reports the percent change in mean Social Security benefits in 2030 for five groups by age in 2030: 50–61, 62–64, 65–69, 70–79 and 80 and older. We compute the change as a weighted ratio of the reform and baseline means. We measure the model’s uncertainty two ways. The draw standard deviation captures the variation across the 20 projections. For the half-sample floor, we split the sample into two halves that share no person and no family unit, take the mean absolute difference between them, and average it over five random splits. The floor shows how far the estimate moves from sampling alone. The comparison lane digitizes DYNASIM3’s projection values from a figure, so each value carries a digitization interval and no standard error.

5.2.1 Exercise 1: a one-point COLA cut

Exercise 1 reduces each annual COLA by one percentage point, starting in 2010. DYNASIM3’s values come from Figure 2 of *Details Matter*, which reports DYNASIM3 run 614 on the 2008 Trustees assumptions (Urban Institute 2010).

Table 3: Exercise 1, a one-point COLA cut: percent change in average benefits in 2030, by age in 2030.

Age in 2030	Microcosm Dynamics	DYNASIM3	Gap (pp)	Half-sample floor
50–61	−7.77	−7.15	−0.62	0.27
62–64	−3.64	−3.79	+0.14	0.35
65–69	−6.26	−6.36	+0.10	0.12
70–79	−12.22	−11.94	−0.28	0.16

Age in 2030	Microcosm Dynamics	DYNASIM3	Gap (pp)	Half-sample floor
80+	−17.92	−18.66	+0.74	0.09

Every age group lands within 0.74 percentage points of DYNASIM3, and the mean absolute gap comes to 0.38 points. The two models order all five groups the same way. The gaps at 62–64 and 65–69 are smaller than the half-sample floor, while those at 50–61, 70–79 and 80 and older exceed it.

The largest gap, 0.74 points, sits in the oldest group. The model’s timing convention applies 21 reduced increases by 2030, so nobody aged 80 or older loses more than 18.56 percent. DYNASIM3’s −18.66 percent lies beyond that bound. A registered alternative that counts the December 2030 increase brings the oldest group within 0.06 points. It also makes every other group’s cut 0.8 to 1.5 points deeper than DYNASIM3’s. No single registered timing convention fits all five groups. Two untested hypotheses could explain the oldest group’s gap: the two models may time COLAs differently for the oldest cohorts, or they may hold different mixes of survivors, including survivors whose cost-of-living increases run from an earlier year. The published figure cannot tell them apart.

The 50–61 group consists almost entirely of disabled workers, and the model’s cut there runs 0.62 points deeper. Two untested hypotheses could account for it: how long the disabled workers on the rolls at the projection’s start have already received benefits, and the 2008-vintage award and termination rates.

Exercise 1 is a weak test of behavioral dynamics: once we fix the year from which each benefit’s cost-of-living increases run, the reform acts mechanically on benefits in payment. The registration lists six alternatives, each changing one convention. An alternative that starts the reduced increases at each beneficiary’s own entitlement year widens the mean gap to 1.47 points, and one built on the 2009 PSID wave gives a mean gap of 0.37.

5.2.2 Exercise 3: a full retirement age of 68

Exercise 3 raises the full retirement age gradually from 2010 until it reaches 68 for people turning 62 in 2022 or later. DYNASIM3’s values come from the “Increase FRA to 68” bars in the same figure (Urban Institute 2010). The test applies the reform to each of exercise 1’s 20 projected paths and holds claiming ages fixed at their baseline values.

Table 4: Exercise 3, a full retirement age of 68: percent change in average benefits in 2030, by age in 2030.

Age in 2030	Microcosm Dynamics	DYNASIM3	Gap (pp)	Half-sample floor
50–61	−0.00	0.00 (reading)	−0.00	0.00
62–64	−4.73	−5.12	+0.39	0.09
65–69	−5.68	−9.90	+4.22	0.11
70–79	−5.24	−4.15	−1.09	0.07
80+	−0.75	−0.60	−0.14	0.11

At 65–69, DYNASIM3’s cut is 1.74 times the model’s, a gap of 4.22 points. That one group carries 72 percent of the mean absolute gap of 1.17 points. Nine of the ten pairwise orderings agree. DYNASIM3’s 50–61 bar shows no visible height, so its zero is a digitizer’s reading, and the drawing’s resolution of about 0.09 points bounds that reading. The model’s estimate is consistent with zero at that resolution.

The 65–69 group covers the 1961–1965 birth cohorts, whose full retirement age rises by 12 months. At fixed claim ages, that rise cuts a retired worker’s benefit by 6.25 to 7.69 percent and a spouse’s excess by at most 10 percent, and it leaves a disabled worker’s benefit where it was. To reach DYNASIM3’s 9.90 percent through those factors, the group would have to consist almost entirely of spouses’ excesses starting at 65. In the model, disabled and converted-disabled benefits make up 17.8 percent of the group’s baseline dollars. DYNASIM3’s 65–69 bar therefore includes more than the age-factor cut at fixed claim ages.

Three untested hypotheses could supply the rest. The first is a retirement earnings test extended to 68, which would withhold benefits from people still working. The model pays gross benefits, applies no earnings test and draws no earnings after 2010, so testing this hypothesis requires forward earnings. The second is a claiming response that shifts claims later. The third is an average over all living people in place of recipients alone, which counts a postponed claimant as zero. A diagnostic that applies that averaging to the model’s own claiming alternative moves the 65–69 change to -8.68 percent.

The registration lists eight alternatives, which vary the PSID wave, the phase-in schedule, survivor rules, a claiming response, benefit types and the statistic. None comes within 3.9 points of DYNASIM3 at 65–69.

Exercise 3’s first registration, number 14 on issue #42, stopped before scoring. The test runner’s membership guard, which then halted any run in which the baseline and the reform counted different people as beneficiaries, ended it. By then the run had computed the primary row in memory. It had printed and recorded no value, no artifact existed, and the seal stayed closed. Registration 15 carried the same forecast and produced the scored run, after a reviewed change that makes the guard halt only on membership differences it does not explain.

5.3 The static calculations

Exercises 2 and 4 apply each reform to incomes the PSID observed and count who it affects. They project nothing and model no behavioral response. The model’s figures describe realized incomes, while DYNASIM3’s come from projections its authors published in 2004 for exercise 2 and in 2006 for exercise 4 (Butrica and Uccello 2004; Favreault et al. 2006). A gap in either test can reflect the population’s vintage as well as the two models’ rules.

5.3.1 Exercise 2: a uniform 13 percent benefit cut

Exercise 2 cuts every Social Security benefit by 13 percent from 2004. DYNASIM3’s values come from Tables 19 and 21 of Butrica and Uccello’s *How Will Boomers Fare at Retirement?*, which report what the authors call adjusted poverty at age 67 for the 1936–45 birth cohort, with and without the cut (Butrica and Uccello 2004). The test reports the change in the poverty rate in percentage points. The model’s sample holds 483 people whom the PSID observes at exactly age 67, born in 1937, 1939, 1941, 1943 and 1945. We attach each person to their family’s income and poverty threshold. The report prints poverty levels as whole percentages, so each printed change carries an open rounding interval of ± 1 percentage point.

Table 5: Exercise 2, a uniform 13 percent benefit cut: change in adjusted poverty at age 67, in percentage points.

Group	n	Microcosm Dynamics	DYNASIM3	Gap (pp)
All	483	+0.47	+2	−1.54
Women	249	+0.35	+3	−2.65
Men	234	+0.60	+1	−0.40
Married	331	+0.60	+1	−0.40

Group	n	Microcosm Dynamics	DYNASIM3	Gap (pp)
Widowed	44	+0.45	+2	−1.55
Divorced	60	0	+5	−5.00
Never married	14	0	+5	−5.00

The cut raises adjusted poverty by 0.47 percentage points in the model and by 2 points in DYNASIM3. The model’s change comes out smaller in all seven groups. Men and married people fall inside DYNASIM3’s rounding interval, and the other five groups fall outside it. Only four sample members cross the poverty line, and each average-weight observation moves the headline by 0.21 points. No divorced or never-married member crosses, so both groups change by zero. Without a crosser, we cannot estimate the uncertainty of either group’s change, however large the group. The model’s poverty levels run lower than DYNASIM3’s in six of seven groups before the cut and in all seven after it.

The registration lists ten alternatives, each changing one input. They vary the birth years covered, the SSI response, the income unit, asset income, employer pension balances, the poverty standard, the life table and the threshold schedule. Five leave every poverty outcome where it was, and none brings the headline inside DYNASIM3’s interval. The closest covers all ten birth years and gives +0.62. The untested hypotheses include thinner PSID coverage of low-income unmarried people and whole-family income standing in for an individual’s resources. They also include income sources the report does not name and self-reported benefits that may be net of Medicare premiums. A fifth is timing: the model uses incomes realized from 2004 to 2012, while DYNASIM3 projects from the 2002 Trustees assumptions.

5.3.2 Exercise 4: a minimum benefit

Exercise 4 adds a minimum benefit tied to years of work. Its four versions combine price or wage indexing with a standard or generous schedule, and an across-the-board cut finances each one. They are options 2, 3, 4 and 5, and we take option 2 as the headline. DYNASIM3’s values come from Table 6 of Favreault, Mermin and Steuerle’s *Minimum Benefits in Social Security*, which reports the 2025 shares of beneficiaries aged 62 and older who receive the minimum, for everyone, men and women (Favreault et al. 2006). The model’s sample holds 2,135 Social Security recipients in the PSID’s 2023 wave, with incomes for 2022. We count a person when their benefit rests on the minimum, through their own record or through a linked spouse’s or survivor’s benefit.

Table 6: Exercise 4, a minimum benefit: percent of beneficiaries aged 62 and older who receive the minimum.

Option	Group	Microcosm Dynamics	DYNASIM3	Gap (pp)
2 (headline)	All	7.75	6.3	+1.45
2	Men	2.36	4.8	−2.44
2	Women	11.67	7.6	+4.07
3	All	11.12	11.3	−0.18
3	Men	3.23	8.7	−5.47
3	Women	16.86	13.4	+3.46
4	All	24.56	18.7	+5.86
4	Men	13.42	14.4	−0.98
4	Women	32.66	22.0	+10.66
5	All	31.47	29.3	+2.17
5	Men	19.24	24.4	−5.16
5	Women	40.36	33.2	+7.16

In every option, the model’s share runs lower than DYNASIM3’s for men and higher for women. The rows for everyone mask the split by sex: under option 3 the model lands 0.18 points from DYNASIM3 for everyone, yet 5.47 points under it for men and 3.46 points over it for women. Both models rank the four options the same way, and both give women a higher share than men in every option. The mean absolute gap across the 12 cells comes to 4.09 points. The untested hypotheses include the difference between a realized 2022 population and a 2025 projection, the source of each person’s PIA, the completeness of PSID earnings histories, and records whose benefits began as disability benefits. They also include 60 auxiliary beneficiaries whom the model cannot link to any worker.

In the primary row, we build each person’s PIA from the earnings history. A registered alternative that takes the PIA from the benefit the person receives lowers the mean gap to 1.60 points. That alternative cannot replace the primary row, whose mean gap of 4.09 points remains the test’s result.

5.4 What the four tests show and what they cannot

Under the COLA cut, the model lands within 0.74 points of DYNASIM3 in every age group. In the full-retirement-age test, the two models differ by 4.22 points in one age group, 65–69. The model’s poverty response to the 13 percent cut comes out smaller in every group, and in every option the model puts a smaller share of men and a larger share of women on the minimum benefit than DYNASIM3 does. We score the blind forecasts separately from the comparisons. The forecast intervals contain every projection estimate, and in each projection test one gap falls outside its interval. In the static tests, the forecast intervals exclude the model’s own values more often than DYNASIM3’s.

Each gap measures a disagreement between two models. No test covers DYNASIM4, the current generation of DYNASIM, so the tests establish no parity with it. The tests do not certify the model. Exercises 1, 3 and 4 compute benefits with Python rules, and exercise 2 cuts reported benefits directly. None of the four runs these rules in the Axiom rules engine. We plan to run the full benefit rules in Axiom as separate work (Section 6). The static tests set observed incomes against DYNASIM3’s projections and never touch the projection engine, while the projection tests exercise that engine without forward earnings, a retirement earnings test or behavioral claiming. Because neither the COLA cut nor the higher retirement age is current law, no observed outcome can check either model.

The registered alternatives measure how sensitive the model is to its own conventions and reveal nothing about how DYNASIM3 computes its results. Each explanation we offer for the gaps stays a hypothesis until a registered test addresses it.

6 What remains to build

Forward earnings are the largest piece the projection tests still lack. The forward earnings law passes its own gate, m6, and no projection test uses it yet. To compare the model with DYNASIM3’s reform results, we need projections that begin in 2010, end in 2030 and turn on people aged 60 to 69. The law does not yet reach those people in their sixties. Ages 60 to 64 sit inside its fitted range of 25 to 64, but the projection cohort starts in 2010, four years before the law’s 2014 boundary, and above 64 the law has no fitted distribution. We plan first to connect forward earnings to the projection tests, and then to build the benefit mechanics that depend on earnings. Each step is a design that joins the model only after it passes a dedicated registered gate, and none has joined yet. Two pieces already exist: the retirement earnings test parameters, and the code for a planned rerun of both projection tests, which we have merged into the repository.

Forward earnings in the projection. We plan to carry the earnings state of a 2010 cohort forward to 2014, the law’s fitted boundary. A new transport gate would score those carried-forward earnings states against the PSID’s 2012 and 2014 waves. We already used those two waves to select the model, and the gate’s registration would say so. We would also fit a second version of the law on a wider

age range, 25 to 69, with one gate for each of three age bands: 62–64, 65–66 and 67–69. Any 2030 claim that rests on forward earnings would need evidence from a planned backtest that uses 2006 as its boundary year and spans the PSID waves from 2006 to 2022. That backtest could not support a 2030 claim alone, because those same waves already shaped our modeling choices. We would record three separate monthly series from 2010 through 2030: labor earnings, covered earnings and the earnings that count under the retirement earnings test.

The retirement earnings test. The earnings test’s parameters agree with SSA’s published history. No projection applies them yet, and we plan to apply them next. An annual ledger would track entitlement, provisional suspension, final deduction and the grace year, and we would validate it against SSA’s worked examples. One separate module would make the adjustment at full retirement age for the months in which the earnings test withheld benefits. Another would recompute benefits when later earnings arrive. One of three untested hypotheses for exercise 3’s largest gap, 4.22 percentage points at ages 65 to 69, is an earnings test extended to age 68. Once we combine the earnings-test machinery with fixed claiming, we could test that hypothesis for the first time.

Gross benefits and dual entitlement. A gross-benefit layer would add the family maximum, age reductions, dual-entitlement reductions and the monthly family payment. We would check it against statutory worked examples, and against Axiom only where a pinned rule exists. A separate rerun, which we plan to register, would repeat both projection tests, exercises 1 and 3, with statutory computation years for disability benefits and with dual entitlement for disability and spouse benefits. We have merged its code into the repository, and so far we have tested it only on invented data. It has no registration yet, and because it comes after the published results, we label it post hoc.

Claiming. The engine draws claim ages from a fixed distribution, and no claim age responds to a reform. We plan to replace that distribution with a monthly claiming hazard, which we would estimate on the insured population that has not yet claimed. We would also add a claiming response identified from policy variation, but only if the data we can access permit a credible design. Even then, to model a claiming response to a full retirement age of 68, we would have to extrapolate past every policy anyone has observed.

Family and population in the projection. The family models pass their gates, 2, 2b and 2c, but none runs inside a projection test yet. We plan to combine the gated marriage and divorce models with the widowhood that a spouse’s death already triggers in the projection tests. We would redraw initial states and declare which populations the projection covers. A registered projection gate would score the result, and a joint validation would check the earnings, claiming, disability, payment and survivor pathways together. In the projection, entrants, both births and immigrants, stay report-only until we admit into the model a way of setting starting states that supports them.

Axiom execution. Axiom reproduces the Python oracle’s AIME for all 7,486 people we tested, and so far it reproduces nothing else. Axiom’s Social Security encodings await acceptance, which must come before the full benefit rules can run in Axiom. We plan to extend Axiom from AIME onward through the PIA, auxiliary benefits and the earnings test.

Wealth, pensions and the long horizon. The model reads PSID wealth and annuitizes it for income measures, and it projects no wealth. The projection tests end in 2030. Any projection we run well past that year carries one more limit: gate m6 certifies nothing about mortality drift.

Held-out targets. We plan two more blind tests under the same protocol as the four existing ones. The first covers the 1946–55 birth cohort from the 2004 report behind exercise 2 (Butrica and Uccello 2004), and its DYNASIM3 values already sit under the existing seal, away from the model’s builders. For this test, we would carry the PSID income and wealth mappings forward into the 2015–2023 waves. The second covers the progressive price indexing option, from the 2010 figure behind exercises 1 and 3 (Urban Institute 2010). The report behind that figure omits the option’s operational formula, so we would need to register conventions for it. We will run no blind test that needs a projected population

until the planned projection gates pass.

7 Related work

The United States Social Security debate relies on a small set of models. The Urban Institute’s DYNASIM projects incomes and retirement outcomes for the population (Favreault and Smith 2004; Favreault et al. 2015), and DYNASIM4 is its current generation (Cosic et al. 2024). MINT, the Social Security Administration’s model, projects the income of future retirees (Smith et al. 2021; Social Security Administration 2024). CBOLT produces the Congressional Budget Office’s long-term projections (O’Harra et al. 2004; Congressional Budget Office 2018, 2024) and contains a microsimulation of individual earnings (Schwabish and Topoleski 2013). The Penn Wharton Budget Model and the Cato model also project Social Security outcomes (Penn Wharton Budget Model 2025; Chanwong 2026). Outside researchers cannot rerun DYNASIM, MINT or CBOLT independently, because their code or administrative records are not public. The Cato model publishes its code and uses public data (Chanwong 2026).

Dynamic microsimulation has a long literature (Harding 1990; Li and O’Donoghue 2013). Outside United States Social Security, the field’s open tools include SimPaths, LIAM2, OpenM++, Destinie 2 and microWELT. SimPaths models the life course (Bronka et al. 2025). LIAM2 (Menten et al. 2014) and OpenM++ (OpenM++ development team 2026) are platforms for building such models. Destinie 2 (Blanchet et al. 2010) and microWELT (Spielauer et al. 2020) model pensions and transfers in aging European populations. Every model of this type has to weight its sample and align simulated outcomes to external totals. Dekkers and Cumpston address weighting (Dekkers and Cumpston 2012), and Li and O’Donoghue address alignment (Li and O’Donoghue 2014). Microcosm Dynamics builds on PolicyEngine’s open cross-section microsimulation and adds a longitudinal layer (PolicyEngine 2026; Ghenis and Woodruff 2024). It differs from the other models in one respect: we publish its code alongside every gate result and every comparison, including every gap the four blind tests find between it and DYNASIM3.

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